

Samuel Rodrigues Silva Costa

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EDUCATION

Otterbein University – Westerville, OH

May 2026

Bachelor of Science in Computer Science, Minor in Data Science

Relevant Courses: Data Structures, Computer Systems, Operating Systems and Computer Networks, Applied Artificial Intelligence, Software Engineering and Practice, Data Science with R, Statistics II, Data Analysis with Python, Database Systems

LANGUAGES

Portuguese (Native) · English (Fluent - C2 Level)

SKILLS

Languages, Tools & Frameworks: Java, Python, C, SQL, Dart, HTML, CSS, JavaScript, Git/GitHub, JUnit, TensorFlow, Ubuntu Linux, JavaFX, Java Swing, Flutter, Supabase, ObjectBox

Concepts: Object-Oriented Programming, Networks and Protocols (TCP/IP, sockets), Data Analysis, Data Manipulation & Visualization, Agile (Scrum), Waterfall, Data Structures and Algorithms, Concurrent Programming and Operating Systems

Soft Skills: Teamwork and collaboration on group projects, technical communication with clients and peers, Problem-solving and logical thinking, Self-directed learning and intellectual curiosity, Empathy, adaptability, and responsibility

PROJECTS & EXTRACURRICULAR

Tutor/Student Scheduler – May 2025

Team Project | Java, JavaFX, CSS | Scrum

- Developed a system that reads Excel (CSV) files, applies graph algorithms to analyze overlapping availability, and displays editable tutor-student pairings via a dynamic JavaFX interface.
- Collaborated with a team of 4 developers using Scrum, including daily meetings and sprint planning for iterative progress.
- Led client meetings to gather requirements, present demos, and incorporate feedback.
- Enabled seamless export of finalized pairings back to Excel, transforming a multi-week manual process into a task of minutes.

Scrabble Game Implementation – December 2024

Team Project | Java, Java Swing | Waterfall

- Implemented core mechanics including tile placement, word validation, and scoring with letter/word multipliers.
- Composed, executed, and produced original background music, enhancing user experience with custom audio.

- Added network functionality for real-time multiplayer gameplay.
- Developed an interactive, polished UI using Java Swing.

Java Asteroids – March 2023

Individual Project | Java, Java Swing

- Recreated the classic arcade game with realistic physics and immersive sound effects.
- Implemented ship movement with acceleration, rotation, and inertia.
- Randomly generated asteroids for unique gameplay experiences.
- Developed scoring, win/lose screens, and interactive interface.
- Applied OOP principles for modular game logic and future expansion.

Graph Analyzer – October 2023

Individual Project | Java

- Built a Java application for graph analysis and manipulation, executing classical graph algorithms based on user input.
- Implemented connectivity checks, minimum spanning trees, and shortest path calculations.
- Added checks for metric graphs and methods to convert graphs to metric.
- Developed exact and approximate solutions to the Traveling Salesman Problem (TSP).

HTTP Server – February 2025

Individual Project | C, Shell Script (Bash), Makefile | Linux (Ubuntu)

- Implemented an HTTP server to handle requests and serve static files.
- Programmed request parsing, file retrieval, and response handling.
- Automated compilation and execution with Bash scripts and Makefile.
- Applied core networking concepts, sockets, and communication protocols.

IDS Server – March 2025

Individual Project | C, Shell Script (Bash), GCC, Makefile | Linux (Ubuntu)

- Developed a C application communicating with a precompiled IDS server via POSIX message queues.
- Implemented client-server communication logic and verification mechanisms for file integrity.
- Automated compilation and execution with Bash and Makefile.

CGI-Based Dynamic Web Server – April 2025

Individual Project | C, Shell Script (Bash), GCC, Makefile | Linux (Ubuntu)

- Developed a dynamic web server in C demonstrating CGI principles and dynamic HTTP content generation.
- Implemented GET/POST request handling and environment variable management for CGI execution.
- Simulated real web server communication between server and CGI scripts.

Ecosystem Simulator – November 2025

Individual Project | Python

- Developed simulator modeling interactions among fish, bears, and algae.
- Implemented population control rules based on energy, reproduction, and environmental balance.
- Applied OOP principles for entity representation and code modularity.

NFA Simulator – October 2025

Individual Project | Python

- Implemented a nondeterministic finite automaton (NFA) simulator with ϵ -transitions.
- Calculated ϵ -closures and reachable states from initial states.
- Modeled NFA structure with lists and dictionaries.
- Used doctests to validate input/output behavior.

D&D Dungeon Master Assistant – Ongoing

Personal Project | Java, Flutter, Dart, Supabase, ObjectBox | Cross-Platform App

- Designing and developing a digital assistant to manage D&D campaigns, automate combat, track players and monsters, and roll dice with history.
- Migrated from Java to Flutter/Dart for cross-platform compatibility.
- Applying modern UI/UX principles for a responsive and intuitive interface.
- Using Supabase and ObjectBox for cloud and local data storage, enabling offline access and synchronization.